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Questions

Q1.

The ionic radius of Al^{3+} is smaller than that of N^{3-}.

This is because Al^{3+} has

- ☒ **A** fewer protons but more electrons than N^{3-}
- ☒ **B** more protons but fewer electrons than N^{3-}
- ☒ **C** more protons than N^{3-} but the same number of electrons as N^{3-}
- ☒ **D** the same number of protons as N^{3-} but fewer electrons

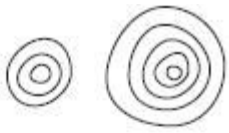
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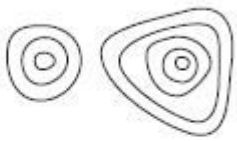
Q2.

Which diagram best represents the electron density map of a hydrogen chloride molecule?

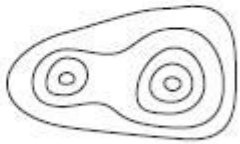
☒ **A**



☒ **B**

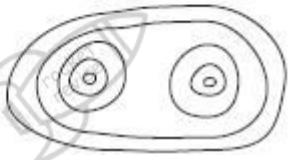


☒ **C**



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☒ **D**



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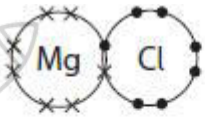
Q3.

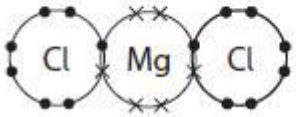
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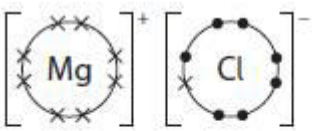
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Which is the dot-and-cross diagram for magnesium chloride?

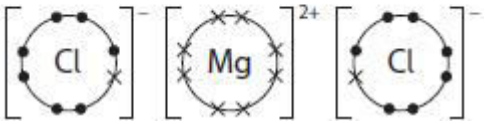
Only outer shell electrons are shown.

☐ **A** 

☐ **B** 

☐ **C** 

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☐ **D** 

(Total for question = 1 mark)

Q4.

Which ion has the greatest polarising power?

☐ **A** Cl^-

☐ **B** Mg^{2+}

☐ **C** Na^+

☐ **D** S^{2-}

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(Total for question = 1 mark)

Q5.

Some ionic radii are shown.

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Ion	Ionic radius / nm
Na^+	0.102
K^+	0.138
F^-	0.133
Cl^-	0.180

Which compound has the strongest ionic bonding?

(1)

- ☒ **A** sodium fluoride
- ☒ **B** sodium chloride
- ☒ **C** potassium fluoride
- ☒ **D** potassium chloride

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(Total for question = 1 mark)

Q6.

In which pair are the ions isoelectronic?

(1)

- ☒ **A** Ca^{2+} and S^{2-}
- ☒ **B** K^+ and Br^-
- ☒ **C** Li^+ and F^-
- ☒ **D** Mg^{2+} and Cl^-

(Total for question = 1 mark)

Q7.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which pair of ions gives the strongest ionic bonding?

- ☒ **A** Na^+ and F^-
- ☒ **B** K^+ and Br^-
- ☒ **C** Mg^{2+} and O^{2-}
- ☒ **D** Ca^{2+} and S^{2-}

(Total for question = 1 mark)

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Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which statement is **not** true for sodium chloride?

- ☒ **A** sodium chloride conducts electricity in aqueous solution
- ☒ **B** sodium chloride conducts electricity when molten
- ☒ **C** sodium chloride has a molecular structure
- ☒ **D** sodium chloride has a giant structure

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(Total for question = 1 mark)

Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which pair of ions will form the compound with the most covalent character?

- ☒ **A** Li^+ and I^-
- ☒ **B** Na^+ and Br^-
- ☒ **C** K^+ and Cl^-
- ☒ **D** Rb^+ and F^-

(Total for question = 1 mark)

Q10.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these ions has the smallest ionic radius?

(1)

- ☒ **A** Al^{3+}
- ☒ **B** Ga^{3+}
- ☒ **C** Mg^{2+}
- ☒ **D** F^-

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(Total for question = 1 mark)

Q11.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these ions has the **greatest** ionic radius?

☐ **A** N^{3-}

☐ **B** F^{-}

☐ **C** Na^{+}

☐ **D** Al^{3+}

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(Total for question = 1 mark)

Q12.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A p-block element in **Period 3** of the Periodic Table reacts to form an ionic compound.

What could be the electronic configuration of the **ion** formed by this element?

☐ **A** $1s^2 2s^2 2p^6 3s^2$

☐ **B** $1s^2 2s^2 2p^6 3s^2 3p^6$

☐ **C** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$

☐ **D** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

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(Total for question = 1 mark)

Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Which ion is the most polarisable?

- ☒ **A** Mg^{2+}
- ☒ **B** Ca^{2+}
- ☒ **C** Cl^{-}
- ☒ **D** I^{-}

(Total for question = 1 mark)

Q14.

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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these ions has the **greatest** polarising power?

- ☒ **A** S^{2-}
- ☒ **B** Cl^{-}
- ☒ **C** K^{+}
- ☒ **D** Ca^{2+}

(Total for question = 1 mark)

Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these isoelectronic ions has the **largest** radius?

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- ☒ **A** Na^{+}
- ☒ **B** Mg^{2+}
- ☒ **C** O^{2-}
- ☒ **D** F^{-}

(Total for question = 1 mark)

Q16.

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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

In which series are the ions in order of **decreasing** ionic radius?

- ☒ **A** $Al^{3+} > Mg^{2+} > Na^{+}$
- ☒ **B** $Li^{+} > Na^{+} > K^{+}$
- ☒ **C** $N^{3-} > O^{2-} > F^{-}$
- ☒ **D** $O^{2-} > S^{2-} > Se^{2-}$

(Total for question = 1 mark)

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Q17.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which properties of an **anion** result in it being most easily polarised?

- ☒ **A** large radius and large charge
- ☒ **B** large radius and small charge
- ☒ **C** small radius and small charge
- ☒ **D** small radius and large charge

(Total for question = 1 mark)

Q18.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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A drop of an aqueous solution of green copper(II) chromate(VI) is placed in the centre of a strip of damp filter paper.
The ends of the filter paper are connected to a DC power supply.

What is observed after a few minutes?

- ☒ **A** a green colour has moved to the negative end
- ☒ **B** a green colour has moved to the positive end
- ☒ **C** a yellow colour has moved to the positive end and a blue colour to the negative end
- ☒ **D** a blue colour has moved to the positive end and a yellow colour to the negative end

(Total for question = 1 mark)

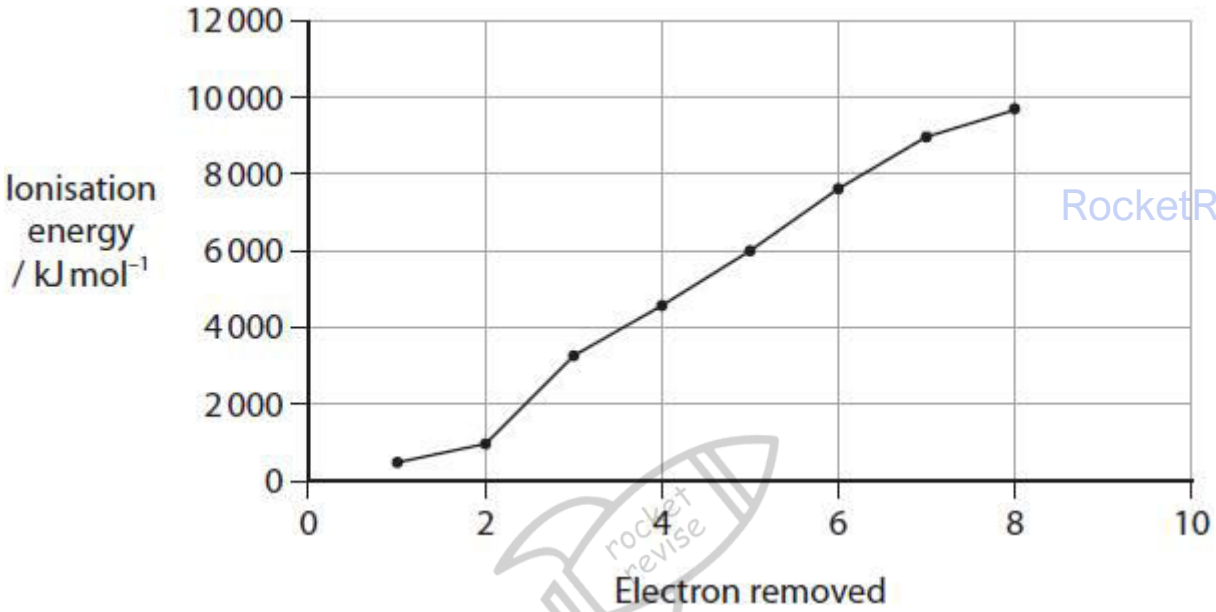
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Q19.

This question is about barium and barium compounds.

(a) The graph shows the first eight ionisation energies of barium.



(i) Write an equation, including state symbols, for the **third** ionisation energy of barium.

(1)

(ii) State how the graph confirms that barium is in Group 2 in the Periodic Table.

(1)

.....

.....

.....

.....

(b) An acidified aqueous solution of barium chloride, $BaCl_2$, is used to test for sulfate ions.

(i) Draw a dot-and-cross diagram for barium chloride.
Show the outer electrons only.

(2)

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(ii) Calculate the mass of barium chloride needed to form 150 cm^3 of $BaCl_2(aq)$ with a concentration of 0.200 mol dm^{-3}.
Give your answer to an appropriate number of significant figures.

(2)

(iii) The barium chloride solution is acidified to prevent carbonate ions giving a false positive result.
Write the **ionic** equation for the reaction of hydrochloric acid with carbonate ions. State symbols are not required.

(1)

(c) Give a reason why solid barium chloride does not conduct electricity.

(1)

.....

.....

(Total for question = 8 marks)

Q20.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which properties of a **cation** result in the greatest polarising power?

- ☒ **A** large radius and large charge
- ☒ **B** large radius and small charge
- ☒ **C** small radius and small charge
- ☒ **D** small radius and large charge

(Total for question = 1 mark)

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Q21.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which row in the table shows the correct forces in a crystal of lithium iodide?

	Attractive forces between ions with opposite charges	Repulsive forces between ions with like charges	Some covalent bonding forces
☐ **A**	✓	X	X
☐ **B**	✓	✓	X
☐ **C**	✓	X	✓
☐ **D**	✓	✓	✓

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Use this space for any rough working. Anything you write in this space will gain no credit.

(Total for question = 1 mark)

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Q22.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these isoelectronic ions has the smallest ionic radius?

- ☐ **A** N^{3-}
- ☐ **B** F^{-}
- ☐ **C** Na^{+}
- ☐ **D** Al^{3+}

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(Total for question = 1 mark)

Q23.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these compounds would you expect to have the highest melting temperature?

- ☒ **A** NaCl
- ☒ **B** NaF
- ☒ **C** KCl
- ☒ **D** KF

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(Total for question = 1 mark)

Q24.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Element **X** is in Group 2 of the Periodic Table and element **Y** is in Group 7. **X** and **Y** are not the symbols of the elements.

(a) What is the formula of the compound formed from **X** and **Y**?

(1)

- ☒ **A** **XY**
- ☒ **B** **X_2Y**
- ☒ **C** **XY_2**
- ☒ **D** **X_2Y_2**

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(b) Under what conditions does the compound formed from **X** and **Y** conduct electricity?

(1)

- ☒ **A** in the solid state and in the liquid state and in aqueous solution
- ☒ **B** in the solid state and in aqueous solution only
- ☒ **C** in the solid state and in the liquid state only
- ☒ **D** in the liquid state and in aqueous solution only

(Total for question = 2 marks)

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Q25.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which has the smallest ionic radius?

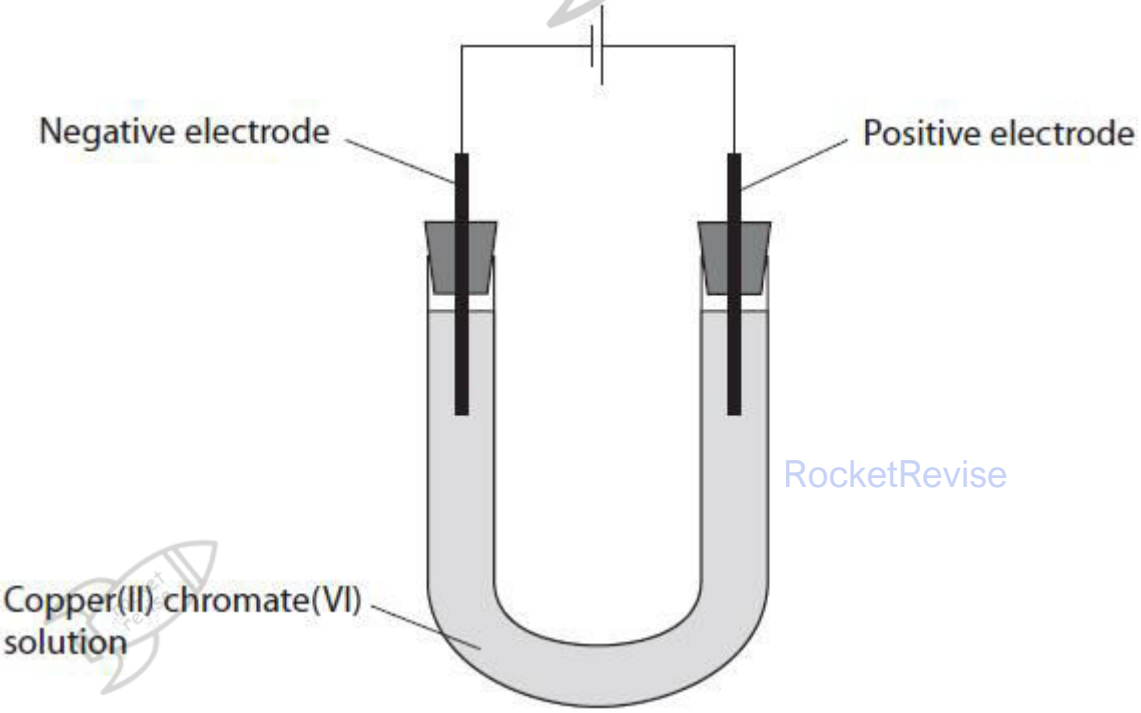
- ☒ **A** F^-
- ☒ **B** Mg^{2+}
- ☒ **C** Na^+
- ☒ **D** O^{2-}

(Total for question = 1 mark)

Q26.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A direct electrical current was passed through a green solution of copper(II) chromate(VI) in the apparatus shown.



Which colours would be seen at each electrode after several minutes?

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	Negative electrode	Positive electrode
☐ A	blue	green
☐ B	green	blue
☐ C	blue	yellow
☐ D	yellow	blue

(Total for question = 1 mark)

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Q27.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which solution contains the **smallest** number of ions?

- ☐ **A** 500 cm^3 of 0.06 mol dm^{-3} $Ca(NO_3)_2(aq)$
- ☐ **B** 500 cm^3 of 0.09 mol dm^{-3} $KI(aq)$
- ☐ **C** 250 cm^3 of 0.12 mol dm^{-3} $BaCl_2(aq)$
- ☐ **D** 250 cm^3 of 0.09 mol dm^{-3} $(NH_4)_2SO_4(aq)$

(Total for question = 1 mark)

Q28.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Which compound has the greatest covalent character?

- ☐ **A** $MgBr_2$
- ☐ **B** MgF_2
- ☐ **C** NaBr
- ☐ **D** NaF

(Total for question = 1 mark)

Q29.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which combination of ions would be expected to form the compound with the **highest** melting temperature?

- ☐ **A** Li^+ and Br^-
- ☐ **B** Mg^{2+} and O^{2-}
- ☐ **C** Sr^{2+} and S^{2-}
- ☐ **D** Rb^+ and Cl^-

(Total for question = 1 mark)

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Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A piece of filter paper is soaked in water and attached to a microscope slide.

A few crystals of green copper(II) chromate(VI) are placed in the centre of the filter paper.

The filter paper is connected to a DC supply of 20 V.

What colours are observed on the paper after a few minutes?



☐	**A**	blue	yellow
☐	**B**	green	blue
☐	**C**	yellow	green
☐	**D**	yellow	blue

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(Total for question = 1 mark)

Q31.

The radii of ions of some successive elements in the Periodic Table are shown.

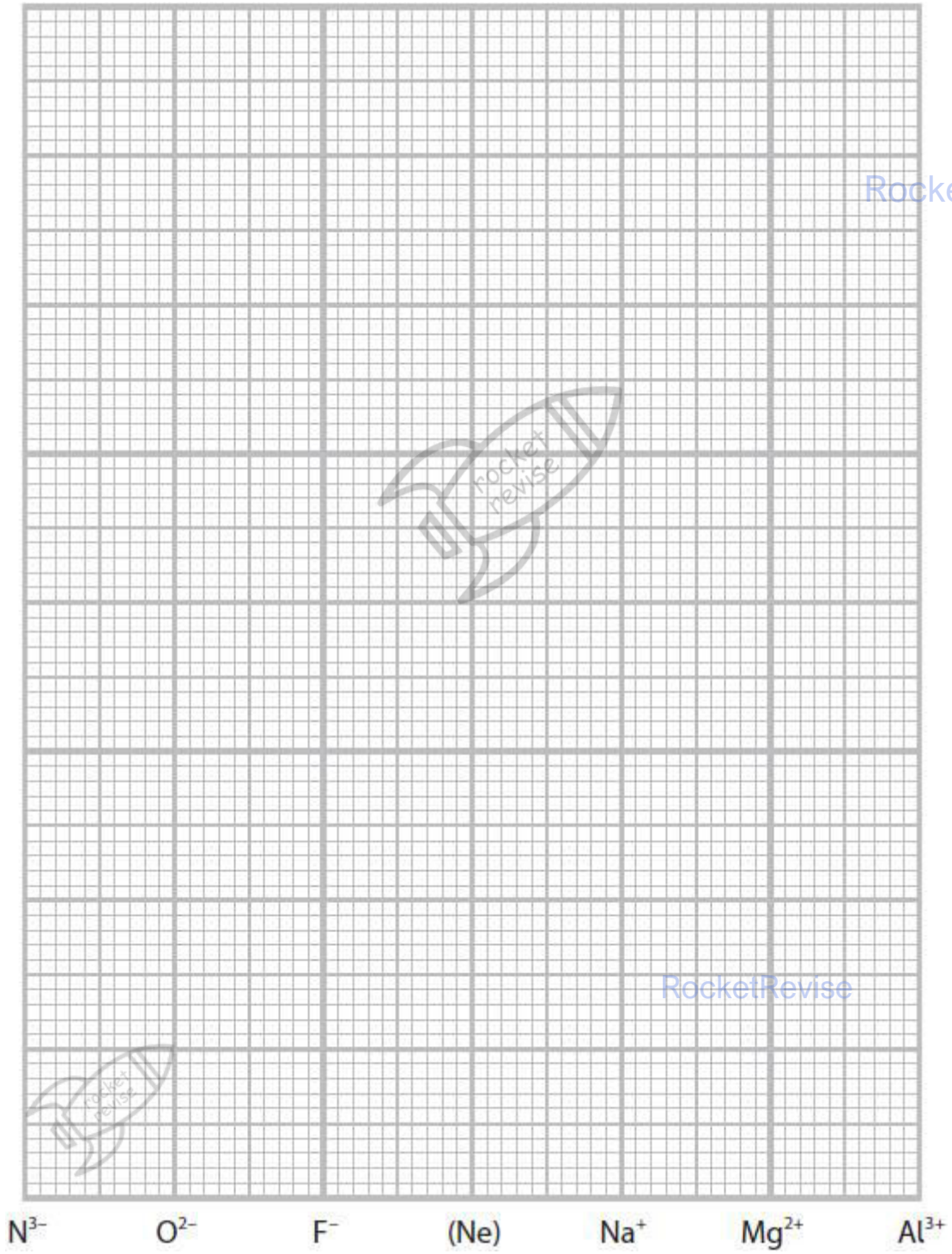
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Ion	N^{3-}	O^{2-}	F^-	Na^+	Mg^{2+}	Al^{3+}
Ionic radius / pm	171	140	133	102		54

(a) (i) Plot a graph of the data.

(2)



(ii) Predict the radius of the magnesium ion using a straight line of best fit on your graph.

(2)

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(iii) Explain the trend in the ionic radii from the nitride ion to the aluminium ion.

(3)

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(b) Sodium and fluorine react to form a compound.

(i) State the structure and bonding of sodium fluoride.

(1)

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(ii) Explain how the electrical conductivity of sodium fluoride provides evidence for the existence of ions.

(3)

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(iii) Explain why the fluoride ion is difficult to polarise.

(2)

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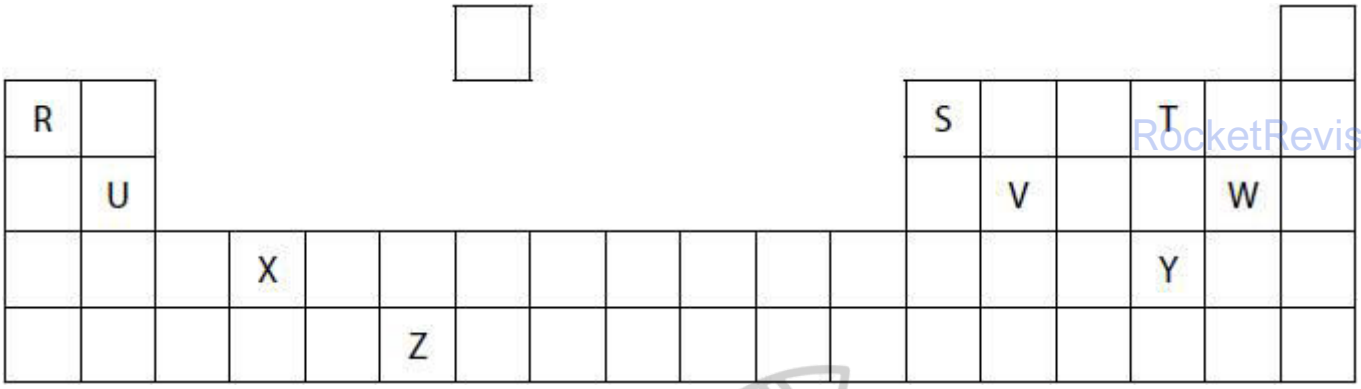
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(Total for question = 13 marks)

Q32.

An outline of part of the Periodic Table is shown.
The letters are not the usual symbols of the elements.



(a) Which elements are in the s-block of the Periodic Table?

(1)

- ☒ **A** R and U
- ☒ **B** T and Y
- ☒ **C** V and W
- ☒ **D** X and Z

(b) Which element has four occupied quantum shells, with six electrons in the outermost shell?

(1)

- ☒ **A** V
- ☒ **B** X
- ☒ **C** Y
- ☒ **D** Z

(c) In which pair do the ions have the same electronic configuration?

(1)

- ☒ **A** R^+ and T^{2-}
- ☒ **B** T^{2-} and Y^{2-}
- ☒ **C** U^{2+} and T^{2-}
- ☒ **D** U^{2+} and W^-

(Total for question = 3 marks)

Q33.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The first ionisation energies of four successive elements in the Periodic Table are shown.

Element	P	Q	R	S
First ionisation energy / kJ mol^{-1}	1251	1521	419	590

(a) Which element has atoms with a full outer shell of electrons?

- ☒ **A** element P
- ☒ **B** element Q
- ☒ **C** element R
- ☒ **D** element S

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(b) Which element could be X in a gaseous covalent compound with the formula HX?

- ☒ **A** element P
- ☒ **B** element Q
- ☒ **C** element R
- ☒ **D** element S

(c) Which element could be Y in an ionic compound with the formula YF_2?

- ☒ **A** element P
- ☒ **B** element Q
- ☒ **C** element R
- ☒ **D** element S

(d) Which element has atoms with the largest atomic radius?

- ☒ **A** element P
- ☒ **B** element Q
- ☒ **C** element R
- ☒ **D** element S

(Total for question = 4 marks)

Q34.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which compound shows the greatest degree of polarisation?

- ☒ **A** sodium chloride
- ☒ **B** sodium iodide
- ☒ **C** magnesium chloride

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☒ **D** magnesium iodide

(Total for question = 1 mark)

Q35.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about the isoelectronic ions N^{3-}, O^{2-}, F^{-}, Na^{+}, Mg^{2+} and Al^{3+}.

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(a) Which ions are shown in order of **decreasing** ionic radius?

(1)

☒ **A** $Na^{+} > Mg^{2+} > Al^{3+}$
☒ **B** $Mg^{2+} > Na^{+} > Al^{3+}$
☒ **C** $Al^{3+} > Na^{+} > Mg^{2+}$
☒ **D** $Al^{3+} > Mg^{2+} > Na^{+}$

(b) Which ions are shown in order of **increasing** ionic radius?

(1)

☒ **A** $F^{-} > O^{2-} > N^{3-}$
☒ **B** $F^{-} > N^{3-} > O^{2-}$
☒ **C** $O^{2-} > N^{3-} > F^{-}$
☒ **D** $N^{3-} > O^{2-} > F^{-}$

(Total for question = 2 marks)

Q36.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which ion is isoelectronic with a noble gas atom (Group 0/8)?

☒ **A** H^{+}
☒ **B** O^{-}
☒ **C** Sc^{3+}
☒ **D** Zn^{2+}

(Total for question = 1 mark)

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Q37.

Which species are arranged in order of **increasing** radius?

- ☒ **A** $K^+ < Ar < Cl^- < Br^-$
- ☒ **B** $Cl^- < Ar < K^+ < Br^-$
- ☒ **C** $Br^- < Cl^- < Ar < K^+$
- ☒ **D** $Cl^- < Br^- < Ar < K^+$

(Total for question = 1 mark)

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Q38.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which cation would be the most polarising?

	Radius	Charge
☒ **A**	small	large
☒ **B**	small	small
☒ **C**	large	small
☒ **D**	large	large

(Total for question = 1 mark)

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